

End-to-End Lifecycle of a Card Transaction with EMV 3D Secure (3DS 2.x)

1. Frontend Payment Initiation

The payment process starts when the customer selects a product, enters the card details, and clicks **Pay Now**. The merchant website sends the payment request to the **Payment Gateway** using the Payment API.

2. Tokenization

The Payment Gateway performs **tokenization**, where the actual card number (PAN) is replaced with a secure token. This helps protect the customer's card information during the transaction.

3. 3D Secure Authentication

The Payment Gateway initiates **3D Secure (3DS)** authentication through the **3DS Server (ACS)**.

- **Frictionless Flow:** Low-risk transactions are authenticated automatically.
- **Challenge Flow:** High-risk transactions require OTP or biometric verification before continuing.

4. Authorization

After successful authentication, the authorization request is sent through the **Card Network** to the **Issuing Bank**. The bank verifies the card status, available balance, CVV, expiry date, and fraud checks before approving or declining the transaction.

5. Success and Failure Handling

The authorization response is returned to the merchant through the Payment Gateway.

- **Success:** Payment is approved, the order is confirmed, and the receipt is generated.
- **Failure:** Payment is declined, an error message is shown, and the customer can retry or choose another payment method.

Conclusion:

This process ensures secure online payments by combining tokenization, 3D Secure authentication, and bank authorization, reducing fraud while providing a safe and reliable payment experience.